Why am I here?!? - A Motivating Example (Revisited)
MECH 412 - *System Dynamics and Control*

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- How does "adaptive cruise control" work?
- Let the desired velocity as a function of time, the **reference** or **command**, be given by

$$r(t) = \bar{r}\left(1 - \exp(-at)\right), \quad \forall t \geq 0, \quad r(s) = \bar{r}\left(\frac{1}{s} - \frac{1}{s+a}\right),$$

  where $\bar{r} \in \mathbb{R}$ and $0 < a < \infty$.
- *Problem statement*: given a desired velocity profile, have the car's velocity follow the desired velocity profile using the engine and the speedometer.

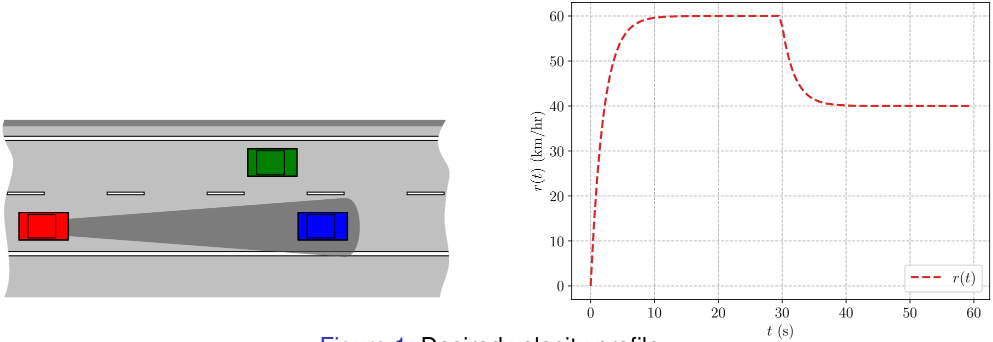

Figure 1: Desired velocity profile.

https://en.wikipedia.org/wiki/Adaptive_cruise_control

Input-Output Relationship

- **Input**: force (due to the engine torque + transmission), $u(t) = f(t)$.
- **Output**: velocity (measured by a speedometer), $y(t) = v(t)$.
- The **relationship** between the **input** and the **output** is given by Newton's 2nd Law (N2L) and the measurement equation,

$$\begin{aligned} m\dot{v}(t) &= \sum_{i=1}^{N} f_i(t), \\ y(t) &= v(t). \end{aligned}$$

- If drag is included, then N2L is written as

$$\begin{aligned} m\dot{v}(t) &= f(t) - bv(t), \quad \forall t \geq 0, \\ y(t) &= v(t), \end{aligned}$$

or

$$\begin{aligned} m\dot{v}(t) + bv(t) &= u(t), \quad \forall t \geq 0, \\ y(t) &= v(t), \end{aligned}$$

where $0 < b < \infty$.

- Taking LTs,

$$\begin{aligned} m\mathcal{L}\left\{\dot{v}(t)\right\} + b\mathcal{L}\left\{v(t)\right\} &= \mathcal{L}\left\{u(t)\right\}, \\ msv(s) - mv_0 + bv(s) &= u(s), \\ (ms+b)v(s) &= u(s) + mv_0, \\ v(s) &= \frac{1}{ms+b}u(s) + \frac{m}{ms+b}v_0, \end{aligned}$$

  where $v_0 = v(0)$.
- Given that $y(s) = v(s)$ it follows that

$$y(s) = \frac{1}{ms+b}u(s) + \frac{m}{ms+b}v_0.$$

- The transfer function from $u(s)$ to $y(s)$ is

$$P(s) = \frac{1}{ms+b} = \frac{y(s)}{u(s)},$$

  and thus

$$y(s) = P(s)u(s) + P(s)mv_0.$$

Problem Statement Revisited

- *Problem statement revisited*: given $r(t)$, have $\lim_{t\to\infty} v(t) = r(t)$ using only $u(t)$ and $y(t)$.

- Recall,
  - **reference/command**: the desired velocity as a function of time, $r(t)$,
  - **input**: force (due to the engine torque + transmission), $u(t) = f(t)$,
  - **output**: velocity (measured by a speedometer), $y(t) = v(t)$.

Feedforward Control

- Consider the *feedforward control law*

$$u(t) = (m\dot{r}(t) + br(t))1_+(t), \quad \text{or} \quad \begin{aligned} u(s) &= msr(s) - mr_0 + br(s) \\ &= (ms+b)r(s) - mr_0 \\ &= C(s)r(s) - mr_0, \end{aligned}$$

  where $C(s) = ms + b = \frac{u(s)}{r(s)}$ and $r_0 = r(0)$.

- Then,

$$\begin{aligned} y(s) &= P(s)u(s) + P(s)mv_0 \\ &= P(s)C(s)r(s) - P(s)mr_0 + P(s)mv_0 \\ &= P(s)C(s)r(s) + P(s)m(v_0 - r_0) \end{aligned}$$

$$\begin{aligned} y(s) &= \left(\frac{1}{ms+b}\right)(ms+b)r(s) + P(s)m(v_0 - r_0) \\ &= r(s) + \frac{m}{ms+b}(v_0 - r_0) \end{aligned}$$

- Notice how feedforward control “cancels out” terms. Is this “canceling out” realistic in the real world? Why or why not?

- The error dynamics are then

$$
\begin{aligned}
e(s) &= r(s) - y(s) \\
&= r(s) - \left(r(s) + \frac{m}{ms+b}(v_0 - r_0) \right) \\
&= \frac{1}{(s + \frac{b}{m})}(-v_0 + r_0).
\end{aligned}
$$

- The time-domain solution is

$$
e(t) = (-v_0 + r_0) \exp\left(-\frac{b}{m} t \right) 1_+(t),
$$

meaning $\lim_{t\to\infty} e(t) = 0$, and thus $\lim_{t\to\infty} v(t) = r(t)$.

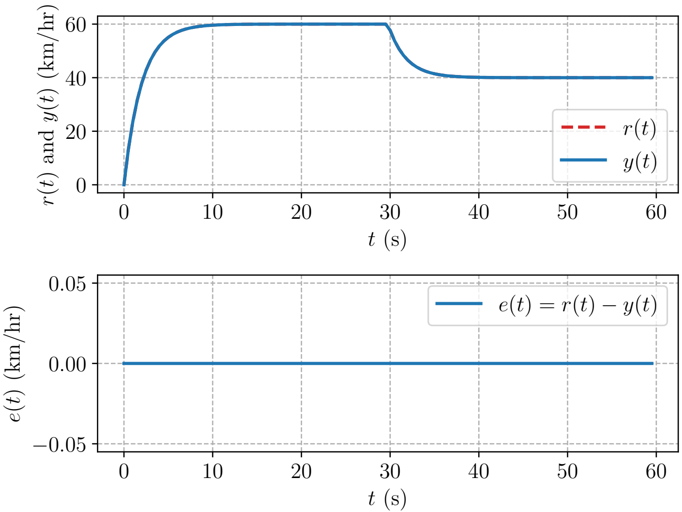

Figure 2: Velocity profile using feedforward control (no model uncertainty).

Clearly $\lim_{t\to\infty} e(t) = 0$, meaning $\lim_{t\to\infty} v(t) = r(t)$. Great!

Discussion

- In reality
  - m, the mass of the car, and
  - b, the drag coefficient,

  are not known *exactly*, meaning there's **model uncertainty**.
- Let m_{a} and b_{a} be the *actual* (i.e., *true*) mass and drag coefficient. Then,

$$y(s) = \frac{1}{m_{\mathrm{a}}s + b_{\mathrm{a}}}u(s) + \frac{m_{\mathrm{a}}}{m_{\mathrm{a}}s + b_{\mathrm{a}}}v_0 = P(s)u(s) + P(s)m_{\mathrm{a}}v_0.$$

- Apply the proposed feedforward control law and see that

$$\begin{aligned} y(s) &= P(s)u(s) + P(s)m_{\mathrm{a}}v_0 \\ &= P(s)\left(C(s)r(s) - mr_0\right) + P(s)m_{\mathrm{a}}v_0 \\ &= P(s)C(s)r(s) - P(s)mr_0 + P(s)m_{\mathrm{a}}v_0 \\ &= \frac{ms + b}{m_{\mathrm{a}}s + b_{\mathrm{a}}}r(s) - \frac{m}{m_{\mathrm{a}}s + b_{\mathrm{a}}}r_0 + \frac{m_{\mathrm{a}}}{(m_{\mathrm{a}}s + b_{\mathrm{a}})}v_0. \end{aligned}$$

- Feedforward control cannot “cancels out” terms in the presence of model uncertainty.

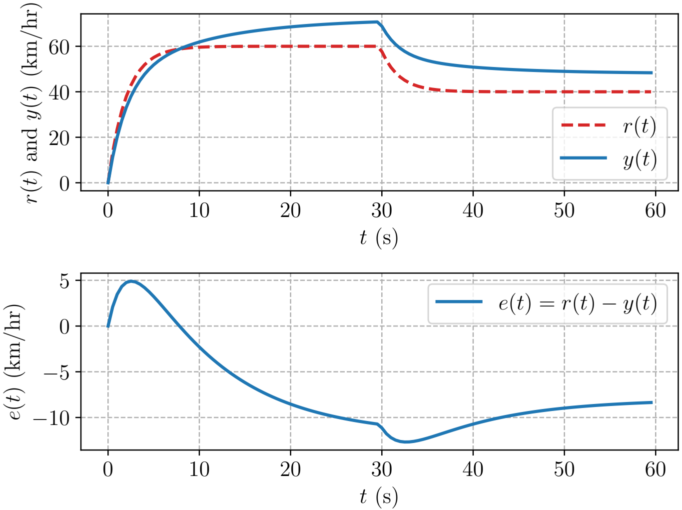

Figure 3: Velocity profile using feedforward control (with model uncertainty).

Clearly $\lim_{t\to\infty} e(t) \neq 0$, meaning $\lim_{t\to\infty} v(t) \neq r(t)$. Boo!

If m and b are not known exactly (and they are *never* known exactly) there will always be error when using a feedforward controller.

Feedback Control

- Consider the *feedback control law*

$$\begin{aligned} u(t) &= k_p(r(t) - y(t)) + k_i \int_0^t (r(\tau) - y(\tau))\mathrm{d}\tau, \\ &= k_p e(t) + k_i \int_0^t e(\tau)\mathrm{d}\tau, \quad \forall t \geq 0, \end{aligned}$$

  where k_p and k_i are constant control parameters (“control gains”) to be tuned/designed, $y(t) = v(t)$, and $e(t) = r(t) - y(t) = r(t) - v(t)$.
- This is a *proportional-integral* (PI) feedback control law.
  - The first term in the PI law is proportional to the error, $e(t)$.
  - The second term in the PI law is proportional to the integral of the error, $\int_0^t e(\tau)\mathrm{d}\tau$.
- Using LTs,

$$u(s) = k_p e(s) + \frac{k_i}{s} e(s) = \left(k_p + \frac{k_i}{s}\right) e(s) = \left(\frac{k_p s + k_i}{s}\right) e(s) = C(s)e(s),$$

  where $C(s) = \frac{k_p s + k_i}{s} = \frac{u(s)}{e(s)}$.

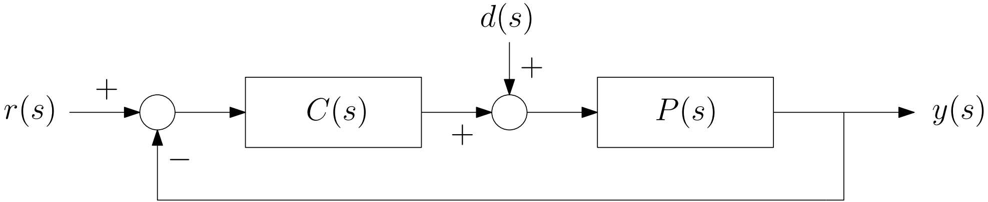

Figure 4: Negative feedback loop (no disturbance is zero, $d(s) = 0$).

- Recalling that

$$y(s) = \frac{1}{ms+b}u(s) + \frac{m}{ms+b}v_0 = P(s)u(s) + P(s)mv_0$$

where $P(s) = \frac{1}{ms+b}$, the error dynamics can be written and simplified via

$$\begin{aligned} e(s) &= r(s) - y(s) \\ &= r(s) - P(s)u(s) - P(s)mv_0 \\ &= r(s) - P(s)C(s)e(s) - P(s)mv_0, \\ (1 + P(s)C(s))e(s) &= r(s) - P(s)mv_0, \\ e(s) &= \frac{1}{1 + P(s)C(s)}r(s) + \frac{P(s)}{1 + P(s)C(s)}(-mv_0). \end{aligned}$$

- The first transfer function can be written as

$$
\begin{aligned}
\frac{1}{1+P(s)C(s)} &= \frac{1}{1+\left(\frac{1}{ms+b}\right)\left(\frac{k_p s+k_i}{s}\right)} \\
&= \frac{s(ms+b)}{ms^2+(b+k_p)s+k_i}.
\end{aligned}
$$

- The second transfer function can be written as

$$
\begin{aligned}
\frac{P(s)}{1+P(s)C(s)} &= \frac{\left(\frac{1}{ms+b}\right)}{1+\left(\frac{1}{ms+b}\right)\left(\frac{k_p s+k_i}{s}\right)} \\
&= \frac{s}{ms^2+(b+k_p)s+k_i}.
\end{aligned}
$$

- Recalling that
$$r(s) = \bar{r}\left(\frac{1}{s} - \frac{1}{s+a}\right),$$
it follows that
$$\begin{aligned}
e(s) &= \frac{1}{1+P(s)C(s)}r(s) + \frac{P(s)}{1+P(s)C(s)}(-mv_0)\\
&= \frac{s(ms+b)}{ms^2+(b+k_p)s+k_i}r(s) - \frac{smv_0}{ms^2+(b+k_p)s+k_i}\\
&= \frac{s(ms+b)}{ms^2+(b+k_p)s+k_i}\left(\frac{1}{s} - \frac{1}{s+a}\right)\bar{r} - \frac{smv_0}{ms^2+(b+k_p)s+k_i}\\
&= \frac{\bar{r}s(ms+b)}{s(ms^2+(b+k_p)s+k_i)} - \frac{\bar{r}s(ms+b)}{(ms^2+(b+k_p)s+k_i)(s+a)} - \frac{smv_0}{ms^2+(b+k_p)s+k_i}\\
&= e_1(s) + e_2(s) + e_3(s)
\end{aligned}$$
where
$$\begin{aligned}
e_1(s) &= \frac{\bar{r}(ms+b)}{ms^2+(b+k_p)s+k_i},\\
e_2(s) &= -\frac{\bar{r}s(ms+b)}{(ms^2+(b+k_p)s+k_i)(s+a)},\\
e_3(s) &= -\frac{smv_0}{ms^2+(b+k_p)s+k_i}.
\end{aligned}$$

- Okay, great …

Q. What do we want to know?

A. If $\lim_{t\to\infty} e(t) = 0$, meaning $\lim_{t\to\infty} v(t) = r(t)$, when the feedback controller

$$u(s) = \left(\frac{k_p s + k_i}{s}\right) e(s)$$

is used.

- What tool do we have in our tool box to evaluate things like $\lim_{t\to\infty} e(t)$?

Final-Value Theorem

Let $f(s) = \mathcal{L}\{f(t)\}$ and assume that $sf(s)$ is analytic in $\mathrm{Re}\ s \geq 0$ (that is, $sf(s)$ has no poles in the CRHP). Then

$$\lim_{t \to \infty} f(t) = \lim_{s \to 0} sf(s).$$

- Where are the poles of $se_i(s)$, $i = 1, 2, 3$?
- Consider $\alpha(s) = ms^2 + (b + k_p)s + k_i$. The roots of $\alpha(s)$ are

$$s_1 = -\frac{(b + k_p)}{2m} + \frac{1}{2m}\sqrt{(b + k_p)^2 - 4mk_i}, \quad s_2 = -\frac{(b + k_p)}{2m} - \frac{1}{2m}\sqrt{(b + k_p)^2 - 4mk_i}$$

- Thus, $\mathrm{Re}\ s_k < 0$, $k = 1, 2$ provided that $b + k_p > 0$ and $k_i > 0$.
- As such none of $se_i(s)$, $i = 1, 2, 3$ have poles in the CRHP provided that $b + k_p > 0$ and $k_i > 0$.
  - Consider $se_2(s)$, which does not have any poles in the CRHP provided that $b + k_p > 0$ and $k_i > 0$. Why?

- Given that none of $se_i(s)$, $i = 1, 2, 3$ have poles in the CRHP, the FVT is applicable.
- It follows that

$$\lim_{t\to\infty} e_1(t) = \lim_{s\to 0} se_1(s) = \lim_{s\to 0} \frac{s\bar{r}(ms+b)}{ms^2 + (b+k_p)s + k_i} = 0,$$

$$\lim_{t\to\infty} e_2(t) = \lim_{s\to 0} se_2(s) = -\lim_{s\to 0} \frac{\bar{r}s^2(ms+b)}{(ms^2 + (b+k_p)s + k_i)(s+a)} = 0,$$

$$\lim_{t\to\infty} e_3(t) = \lim_{s\to 0} se_3(s) = -\lim_{s\to 0} \frac{s^2 m v_0}{ms^2 + (b+k_p)s + k_i} = 0.$$

- Therefore, $\lim_{t\to\infty} e(t) = 0$, meaning $\lim_{t\to\infty} v(t) = r(t)$!

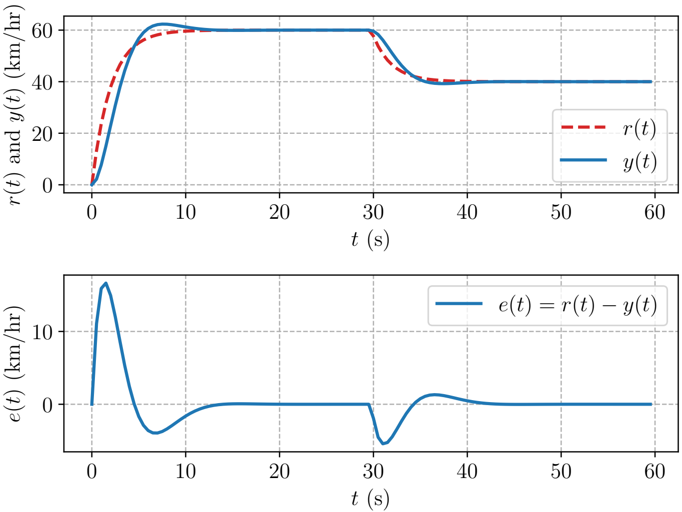

Figure 5: Velocity profile using feedback control (no model uncertainty).

Clearly $\lim_{t\to\infty} e(t) = 0$, meaning $\lim_{t\to\infty} v(t) = r(t)$. Great!

Discussion

- The proposed feedback control law
$$u(t) = k_p(r(t) - y(t)) + k_i \int_0^t (r(\tau) - y(\tau))\mathrm{d}\tau = k_p e(t) + k_i \int_0^t e(\tau)\mathrm{d}\tau, \quad \forall t \geq 0,$$
uses **the measurement** $y(t)$ and does not rely on knowledge of m and b.
- Let m_a and b_a be the *actual* (i.e., *true*) mass and drag coefficient. Tthen,
$$P(s) = \frac{1}{m_\mathrm{a} s + b_\mathrm{a}}.$$
- Repeating the previous analysis, it follows that
$$\begin{aligned} e(s) &= \frac{1}{1 + P(s)C(s)} r(s) + \frac{P(s)}{1 + P(s)C(s)}(-m_\mathrm{a} v_0) \\ &= \frac{\bar{r}(m_\mathrm{a} s + b_\mathrm{a})}{m_\mathrm{a} s^2 + (b_\mathrm{a} + k_p)s + k_i} - \frac{\bar{r} s(m_\mathrm{a} s + b_\mathrm{a})}{(m_\mathrm{a} s^2 + (b_\mathrm{a} + k_p)s + k_i)(s + a)} - \frac{s m_\mathrm{a} v_0}{m_\mathrm{a} s^2 + (b + k_p)s + k_i} \\ &= e_1(s) + e_2(s) + e_3(s) \end{aligned}$$
where
$$e_1(s) = \frac{\bar{r}(m_\mathrm{a} s + b_\mathrm{a})}{m_\mathrm{a} s^2 + (b_\mathrm{a} + k_p)s + k_i}, \quad e_2(s) = -\frac{\bar{r} s(m_\mathrm{a} s + b_\mathrm{a})}{(m_\mathrm{a} s^2 + (b_\mathrm{a} + k_p)s + k_i)(s + a)},$$
$$e_3(s) = -\frac{s m_\mathrm{a} v_0}{m_\mathrm{a} s^2 + (b_\mathrm{a} + k_p)s + k_i}.$$

- Again, none of $se_i(s)$, $i = 1, 2, 3$ have poles in the CRHP provided that $b_\mathrm{a} + k_p > 0$ and $k_i > 0$, and the FVT is applicable.
- It follows that

$$\lim_{t\to\infty} e_1(t) = \lim_{s\to 0} se_1(s) = \lim_{s\to 0} \frac{s\bar{r}(m_\mathrm{a}s + b_\mathrm{a})}{m_\mathrm{a}s^2 + (b_\mathrm{a} + k_p)s + k_i} = 0,$$

$$\lim_{t\to\infty} e_2(t) = \lim_{s\to 0} se_2(s) = -\lim_{s\to 0} \frac{\bar{r}s^2(m_\mathrm{a}s + b_\mathrm{a})}{(m_\mathrm{a}s^2 + (b_\mathrm{a} + k_p)s + k_i)(s + a)} = 0,$$

$$\lim_{t\to\infty} e_3(t) = \lim_{s\to 0} se_3(s) = -\lim_{s\to 0} \frac{s^2 m_\mathrm{a} v_0}{m_\mathrm{a}s^2 + (b_\mathrm{a} + k_p)s + k_i} = 0.$$

- Therefore, $\lim_{t\to\infty} e(t) = 0$, meaning $\lim_{t\to\infty} v(t) = r(t)$, even when m_a and b_a are uncertain!
- In fact, $\lim_{t\to\infty} e(t) = 0$ for any value of m_a and for any $b_\mathrm{a} + k_p > 0$! The feedback controller is robust to a huge amount of parameter variation!
  - For instance, say the worst case drag is $b_\mathrm{a} = 0$. Just have $0 < k_p < \infty$.

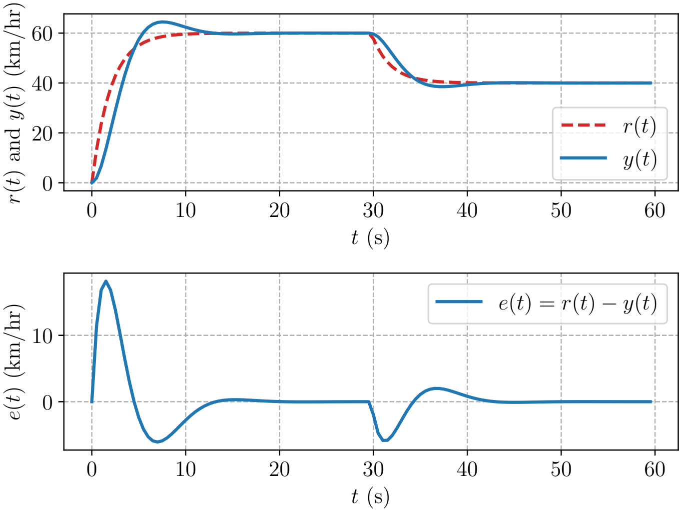

Figure 6: Velocity profile using feedback control (with model uncertainty).

Clearly $\lim_{t\to\infty} e(t) = 0$, meaning $\lim_{t\to\infty} v(t) = r(t)$, even in the presence of **model uncertainty**.

In fact, the response barely changes in the presence of model uncertainty!

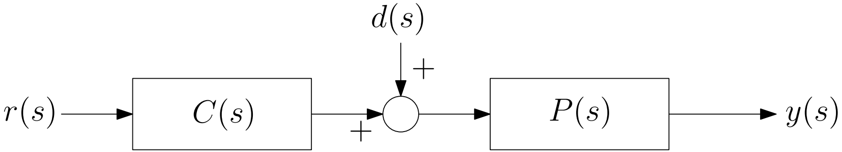

Figure 7: Feedforward control.

Feedforward control

- relies on exact model information, and
- does not use measurement $y(t)$.

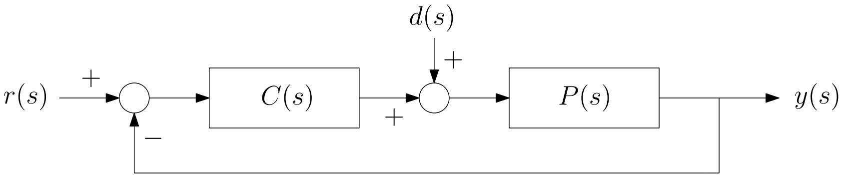

Figure 8: Feedback control.

Feedback control

- does not rely on exact model information, and
- uses measurement $y(t)$.

References

Notes roughly based on [1] and the example found at `http://ctms.engin.umich.edu/CTMS/index.php?example=CruiseControl&section=SystemModeling`.

[1] L. Qiu and K. Zhou, *Introduction to Feedback Control*. Upper Saddle River, NJ: Prentice-Hall, Inc., 2010.